Constants From Balanced Ternary

The Forced Arithmetic of $\{-1, 0, +1\}$

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Abstract

The companion paper *Balanced Ternary by Necessity* established that $\{-1, 0, +1\}$ is the unique minimal integer-valued state space capable of intrinsically representing a directed transition. This paper asks the next question: what mathematical structure appears when that alphabet is subjected to successive completion demands beyond its primitive one-step content?

Starting from $\{-1, 0, +1\}$, we construct a derivation ladder. Some constants arise once independent generators, metric comparison and symmetry-preserving operators are admitted. Others appear only when the substrate is examined through increasingly strong analytical completions, including recurrence, refinement, rotation and summation. The claim of this paper is therefore ordered rather than absolute: the constants are not all primitive contents of the ternary alphabet, but they do appear in a strict sequence once each additional structural demand is made explicit.

No physical assumption is made at any step. Mathematical structure is introduced only when required, and each constant enters at the earliest stage permitted by the available machinery. The result is a disciplined map from a minimal discrete substrate to the hierarchy of irrational and transcendental constants naturally exposed by its admissible completions.

Reading guide

This paper is the third in a sequence.

The philosophical argument for why anything exists at all is given in:

Alan Ball, *On the Necessity of Existence* (Zenodo, 2026), DOI: 10.5281/zenodo.18797375

The formal proof that $\{-1, 0, +1\}$ is the unique minimal state space for directed transitions is given in:

Alan Ball, *Balanced Ternary by Necessity* (Zenodo, 2026), DOI: 10.5281/zenodo.18806015

This paper begins where that proof ends. It takes the established alphabet and asks what happens when you operate on it. Each section derives one or more constants. The derivation chain is sequential: each result depends only on results already established.

This paper is deliberately silent on physical interpretation.

While the reader may notice correspondences;

This paper does not comment on them.

I. The Geometry of the Alphabet

1 The second axis and the quarter-turn

Starting inventory: $\{-1, 0, +1\}$ on a single axis with generator e and transition operators τ_e and τ_e^{-1}.

The question:

Can a second directed transition exist that is independent of the first?

The requirement:

Let f denote a second generator with the same single-step contract as e, but not reducible to $\pm e$ by rescaling or relabelling. We call this requirement *independence*.

Independence alone yields a copy of $\mathbb{Z}^2$. But this by itself does not distinguish a plane from an arbitrary product of two lines.

To obtain an intrinsic notion of "right angle" without importing any physical geometry, we require the existence of a structure-preserving quarter-turn operator J acting on the span of the generators such that

$$J(e) = f, \quad J(f) = -e$$

This implies $J^2(e) = -e$ and $J^2(f) = -f$, hence $J^2 = -1$ on the generator subspace.

The result:

A second independent axis by itself does not force an operator whose square is negation. What forces it is the additional demand that the two-generator extension admit an intrinsic, structure-preserving quarter-turn.

Under that demand, $\sqrt{-1}$ enters as a purely structural object: not as a number imported from elsewhere, but as the unique algebraic witness of a quarter-turn that cannot be realised on a single signed line.

With this operator, the natural integer-coordinate plane is $\mathbb{Z}[i]$ (the Gaussian integers), where i denotes the action of J and $i^2 = -1$.

Inventory update: $\{-1, 0, +1\}$, two independent generators e and f, the operator J with $J^2 = -1$.

$$\boxed{i = \sqrt{-1}}$$

2 The diagonal and $\sqrt{2}$

The question:

What is the distance of the diagonal step $e + f$ on the plane?

The requirement:

To answer this, we need a notion of length. We impose the minimal structural requirement: the existence of an intrinsic quadratic invariant preserved by the quarter-turn J.

Concretely, assume there exists a function $\|\cdot\|$ on the integer span of $\{e, f\}$ such that:

(i) $\|e\| = \|f\|$ (axis symmetry),

(ii) $\|J(v)\| = \|v\|$ (quarter-turn invariance),

(iii) $\|v\|^2$ is a quadratic form compatible with additivity and negation.

These conditions determine, up to an overall scale, the Euclidean form

$$\|ae + bf\|^2 \ \propto \ a^2 + b^2$$

We fix the overall scale by choosing units such that $\|e\| = \|f\| = 1$. With this normalisation, the diagonal step $e + f$ satisfies:

$$\|e + f\|^2 = \|e\|^2 + \|f\|^2 = 1 + 1 = 2 \quad \Rightarrow \quad \|e + f\| = \sqrt{2}\,\|e\|$$

The result:

$\sqrt{2}$ is not an imported constant. It is the first unavoidable irrational that appears once a plane supports two independent unit steps together with an intrinsic quarter-turn preserving a quadratic invariant.

The substrate has asked a question about its own geometry and received an answer it cannot express as a ratio of integers.

$$\boxed{\sqrt{2} = 1.41421356\ldots}$$

3 The third axis and $\sqrt{3}$

The question:

Can a third directed transition exist that is independent of both e and f?

The requirement:

We introduce a third generator g, independent of both e and f, with the same single-step contract. As an additional structural assumption, we require that each coordinate plane admits a structure-preserving quarter-turn symmetry, mirroring the 2D case.

This extends the reachability set from $\mathbb{Z}^2$ to $\mathbb{Z}^3$. The quarter-turn requirement in each coordinate plane yields three operators:

$$J_{ef}^2 = J_{fg}^2 = J_{ge}^2 = -1$$

with actions such as $J_{ef}(e) = f$, $J_{ef}(f) = -e$, and similarly for the other pairs.

The unit-cube diagonal step $e + f + g$ now has:

$$\|e + f + g\|^2 = \|e\|^2 + \|f\|^2 + \|g\|^2 = 3 \quad \Rightarrow \quad \|e + f + g\| = \sqrt{3}\,\|e\|$$

The result:

The unit-cube diagonal of the minimal three-axis integer-coordinate space has length $\sqrt{3}$.

We also note: in three dimensions, the quarter-turn operators J_{ef}, J_{fg}, J_{ge} do not commute under composition. The smallest consistent algebra that accommodates three pairwise plane-rotation quarter-turn generators is quaternionic in character. This paper does not develop the quaternionic structure further; it notes only that it is forced by the extension from two axes to three.

$$\boxed{\sqrt{3} = 1.73205080\ldots}$$

4 The first non-trivial integer-coordinate distance and $\sqrt{5}$

The question:

What integer-coordinate distances exist beyond the axis, the unit-square diagonal, and the unit-cube diagonal?

The requirement:

The integer-coordinate space $\mathbb{Z}^3$ contains all integer-coordinate vectors. The squared distances from the origin are of the form $a^2 + b^2 + c^2$ for integers a, b, c.

The first few distinct squared distances are:

$$1, \quad 2, \quad 3, \quad 4, \quad 5, \quad \ldots$$

Squared distance 4 is simply 2^2: a two-step axis move. It is the first composite. Squared distance 5 is the first that introduces a genuinely new irrational. It is realised by any integer vector of the form $(1, 2, 0)$ or its permutations and sign changes.

$$\|(e + 2f)\|^2 = 1 + 4 = 5 \quad \Rightarrow \quad \|e + 2f\| = \sqrt{5}\,\|e\|$$

The result:

$\sqrt{5}$ is the shortest integer-coordinate distance that cannot be reduced to a multiple of the axis length, the unit-square diagonal, or the unit-cube diagonal.

$$\boxed{\sqrt{5} = 2.23606797\ldots}$$

II. The Self-Referential Constants

5 The golden ratio φ

The question:

What appears when the integer sequence is subjected to a simple, self-referential additive recurrence?

The derivation:

The integers $\mathbb{Z}$ support integer sequences and therefore many possible recurrences. Not all of them are equally informative. Constant, periodic or purely sign-flipping recurrences remain too degenerate to drive a genuine growth law. To obtain a non-trivial self-referential rule using only addition of previously available terms, we choose the recurrence

$$a(n) = a(n-1) + a(n-2)$$

This is the Fibonacci recurrence. It is not claimed here that this choice is the only recurrence the integers permit. It is claimed that it is a highly natural and minimal additive recurrence once one asks for non-degenerate growth built from prior terms alone.

Its characteristic equation is:

$$x^2 = x + 1 \quad \Rightarrow \quad x^2 - x - 1 = 0 \quad \Rightarrow \quad x = \frac{1 \pm \sqrt{5}}{2}$$

The positive root is the golden ratio:

$$\varphi = \frac{1 + \sqrt{5}}{2}$$

This is the asymptotic growth rate of the sequence. It is also the unique positive number satisfying $\varphi = 1 + 1/\varphi$: the ratio that is self-similar under decomposition into itself plus unity.

The result:

φ is not imported. It appears as the growth rate of a chosen but highly natural non-trivial additive recurrence on the integers, and it is built from $\sqrt{5}$, which was itself exposed by the integer-coordinate geometry.

$$\boxed{\varphi = \frac{1 + \sqrt{5}}{2} = 1.61803398\ldots}$$

6 The natural exponential e

The question:

What appears once the unit step is examined through arbitrary refinement and compounded growth?

The derivation:

The substrate has a unit transition of magnitude 1. By itself, this does not yet contain continuous growth. To reach that regime, an additional completion demand must be admitted: the unit step is allowed to be refined into n equal sub-steps, each of magnitude $1/n$, while preserving the same total action.

If growth compounds multiplicatively at each refined sub-step, the accumulated factor after n such steps is

$$\left(1 + \frac{1}{n}\right)^n$$

As $n \to \infty$, this converges:

$$\lim_{n\to\infty} \left(1 + \frac{1}{n}\right)^n = e$$

The point is not that the primitive ternary alphabet is already "doing" infinite subdivision. The point is that, once refinement is admitted as an analytical completion of the unit step, this limit is no longer optional. It is the unique invariant associated with maximal compounding under arbitrary equal subdivision.

The result:

e is the natural base of continuous growth exposed when a discrete unit transition is carried through the completion demand of arbitrary refinement.

$$\boxed{e = 2.71828182\ldots}$$

7 The half-period π

The question:

What constant appears once the previously established objects i and e are carried through a rotation-based analytical completion?

The derivation:

We have established the complex plane $\mathbb{Z}[i]$ and the exponential function e^x. By themselves, these do not yet compel a circle. To reach that regime, an additional completion demand must be admitted: imaginary exponents are allowed, so that the exponential is examined along the purely imaginary direction it.

Under that demand, the standard power-series expansion gives

$$e^{it} = \sum_{k=0}^{\infty} \frac{(it)^k}{k!} = \cos t + i \sin t$$

where $\cos t$ and $\sin t$ are defined by the even and odd parts of the same series. This turns the exponential into a rotation-valued object on the complex plane.

The function e^{it} starts at 1 when $t = 0$. It first returns to the real axis on the negative side when $e^{it} = -1$. The value of t at which this occurs is π.

π is therefore the half-period of unit rotation once the earlier algebraic machinery is carried through this rotational completion. It is not being claimed that raw ternary syntax already contains a physical circle. It is being claimed that, once the complex exponential is admitted, the half-period is no longer optional. It is the unique real number satisfying

$$e^{i\pi} = -1$$

There is a second, corroborating route to the same constant. On the integer grid $\mathbb{Z}^2$, count the number of integer points inside a circle of radius r. In the limit of large r, the ratio of this count to r^2 converges to π. This does not make the circle primitive; it shows that once rotation-invariant comparison is admitted, the same constant reappears through lattice counting.

The result:

π is the half-period of the unit rotation exposed when the already established objects i and e are carried through a rotational analytical completion.

$$\boxed{\pi = 3.14159265\ldots}$$

III. The Convergence

8 Euler's identity

The five quantities 0, 1, e, i, and π have been derived independently:

- 0: the ground state (Paper 2, Constraint G).
- 1: the unit excitation (Paper 2, Constraint T).
- i: the quarter-turn witness (Section 1).
- e: the limit of unit compounding (Section 6).
- π: the half-period of unit rotation (Section 7).

They satisfy:

$$e^{i\pi} + 1 = 0$$

This is not a coincidence and not a construction. It is a constraint: the five objects derived from the substrate are not independent of one another. They are locked together by the algebra that produced them.

The substrate's constants form a closed system.

$$\boxed{e^{i\pi} + 1 = 0}$$

IV. The Information-Theoretic Constants

9 The natural logarithms: $\ln 2$ and $\ln 3$

The question:

Given the natural base e, what is the information content of a distinction on the substrate?

The derivation:

The natural logarithm $\ln x$ is the inverse of the exponential e^x. Having derived e, this inverse is forced.

The most primitive distinction the substrate supports is binary: ground versus not-ground, 0 versus ± 1. The information content of this distinction, measured in natural units, is:

$$\ln 2 = 0.69314718\ldots$$

The full state space $\{-1, 0, +1\}$ is a ternary alphabet. The information content of a single ternary choice is:

$$\ln 3 = 1.09861228\ldots$$

This is the information capacity of one symbol of the substrate. It is not assigned; it is measured, using the exponential base the substrate itself produced.

Derived constants:

$\ln 10$ follows by the product rule. Since $10 = 2 \times 5$:

$$\ln 10 = \ln 2 + \ln 5 = 2.30258509\ldots$$

$$\boxed{\ln 2 = 0.69314718\ldots \qquad \ln 3 = 1.09861228\ldots}$$

V. The Summation Constants

10 $\zeta(2)$: the Basel sum

The question:

What is the sum of inverse squares over the positive integers?

The derivation:

The integers $\mathbb{Z}$ contain the positive integers $1, 2, 3, \ldots$ by repeated composition of the unit step. The sum

$$\zeta(2) = \sum_{n=1}^{\infty} \frac{1}{n^2}$$

is the simplest convergent series formed from inverse powers of the integers' own counting sequence.

Its value was first computed by Euler:

$$\zeta(2) = \frac{\pi^2}{6}$$

π was derived in Section 7. The sum of inverse squares over the integers therefore returns π, squared and scaled. The substrate, summing over its own structure, rediscovers a constant it had already produced by a different route.

$$\boxed{\zeta(2) = \frac{\pi^2}{6} = 1.64493406\ldots}$$

11 $\zeta(3)$: Apéry's constant

The question:

What appears once summation over the positive integers is extended from inverse squares to inverse cubes?

The derivation:

By repeated composition of the unit step, the substrate supports the counting sequence $1, 2, 3, \ldots$. By itself, this does not yet compel a global sum over all inverse cubes. To reach that regime, an additional analytical completion must be admitted: the integer sequence is now treated as the domain of an infinite summation operator.

Under that demand,

$$\zeta(3) = \sum_{n=1}^{\infty} \frac{1}{n^3}$$

Unlike $\zeta(2)$, this sum has no known closed form in terms of π or any other previously derived constant. Apéry proved in 1978 that $\zeta(3)$ is irrational.

The point is not that raw ternary syntax is already evaluating global series in the dark. The point is that, once summation over the integer sequence is admitted as a completion demand, the inverse-cube series yields a constant that appears algebraically independent of the preceding ones and resists reduction to the earlier ladder.

The result:

$\zeta(3)$ is the constant exposed when the already available counting sequence is carried through the stronger analytical completion of inverse- cube summation.

$$\boxed{\zeta(3) = 1.20205690\ldots}$$

12 The Euler–Mascheroni constant γ

The question:

What is the stable gap between discrete summation and continuous accumulation?

The derivation:

The unit step generates the counting sequence $1, 2, 3, \ldots$ and therefore the harmonic partial sums

$$H_n = \sum_{k=1}^{n} \frac{1}{k}$$

Having derived the natural logarithm, we can compare this discrete sum to the continuous accumulation of $1/x$:

$$\int_1^n \frac{dx}{x} = \ln n$$

The difference

$$H_n - \ln n$$

converges as $n \to \infty$ to a constant. This limiting gap is the Euler–Mascheroni constant:

$$\gamma = \lim_{n\to\infty} (H_n - \ln n)$$

Equivalently,

$$\gamma = \int_1^\infty \left(\frac{1}{\lfloor x \rfloor} - \frac{1}{x} \right) dx$$

which makes explicit that γ measures the residue between discrete counting and its smooth logarithmic approximation.

The result:

γ is the constant error term forced by comparing the integers' most primitive divergent sum to the logarithm it asymptotically shadows. It is the simplest measure of how discrete accumulation departs from its continuous limit.

$$\boxed{\gamma = 0.57721566\ldots}$$

13 Catalan's constant G

The question:

What happens when the substrate alternates sign over its own odd inverse squares?

The derivation:

The balanced-ternary substrate has sign as an intrinsic feature. The simplest alternating sum over the odd inverse squares is:

$$G = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)^2} = \frac{1}{1^2} - \frac{1}{3^2} + \frac{1}{5^2} - \frac{1}{7^2} + \cdots$$

This is Catalan's constant. Like $\zeta(3)$, it has no known closed form in terms of elementary constants, and its irrationality remains unproven. It arises naturally in analysis (for example as the Dirichlet beta value $\beta(2)$), and in a variety of integrals and integer-grid model evaluations.

The substrate, exercising its intrinsic sign structure over its own counting sequence, produces a constant that remains opaque to further simplification.

$$\boxed{G = 0.91596559\ldots}$$

14 The lemniscate constant ϖ

The question:

What constant appears once arc-length measurement is extended beyond the circular regime to a simple self-crossing algebraic curve?

The derivation:

The lemniscate of Bernoulli, defined in polar coordinates by $r^2 = \cos 2\theta$, is the simplest algebraic curve that closes on itself with a crossing at the origin. By itself, the previously established ladder does not yet compel elliptic arc-length measurement. To reach that regime, an additional analytical completion must be admitted: the length of a non-circular smooth curve is now treated as a legitimate global invariant.

Under that demand, the total arc length of the lemniscate is

$$L = 2\varpi$$

where ϖ (the lemniscate constant) is given by

$$\varpi = 2\int_0^1 \frac{dt}{\sqrt{1-t^4}}$$

This is a complete elliptic integral. The point is not that raw ternary syntax is already measuring arc lengths of exotic curves. The point is that, once arc-length comparison is extended beyond the circle, the next natural class of invariants is elliptic rather than circular, and the corresponding constant is no longer π but ϖ.

The two constants are related by

$$\varpi = \frac{\Gamma(1/4)^2}{2\sqrt{2\pi}}$$

where Γ is the gamma function, itself the unique smooth extension of the factorial to the reals.

The result:

ϖ is the constant exposed when arc-length measurement is carried from the circular case into the stronger analytical completion of elliptic geometry.

$$\boxed{\varpi = 2.62205755\ldots}$$

15 Summary of derived constants

The following constants have been derived, in order, from the substrate $\{-1, 0, +1\}$ and the successive structural or analytical demands admitted in this paper.

Constant	**Value**	**Appears under**
i	$\sqrt{-1}$	Quarter-turn between independent axes
$\sqrt{2}$	$1.41421\ldots$	Quadratic comparison on the unit square
$\sqrt{3}$	$1.73205\ldots$	Quadratic comparison on the unit cube
$\sqrt{5}$	$2.23606\ldots$	First non-trivial integer-coordinate distance
φ	$1.61803\ldots$	A chosen non-trivial integer recurrence
e	$2.71828\ldots$	Arbitrary refinement with multiplicative compounding
π	$3.14159\ldots$	Rotational completion of the complex exponential
$\ln 2$	$0.69314\ldots$	Binary distinction measured in the base e
$\ln 3$	$1.09861\ldots$	Ternary distinction measured in the base e
$\zeta(2)$	$\pi^2/6$	Inverse-square summation over the integers
$\zeta(3)$	$1.20205\ldots$	Inverse-cube summation over the integers
γ	$0.57721\ldots$	Harmonic–logarithm comparison
G	$0.91596\ldots$	Alternating inverse-square summation
ϖ	$2.62205\ldots$	Elliptic arc-length completion

Every entry in this table appears only once the available inventory is carried through an additional explicit demand. In the earlier sections, those demands were structural: independence, quadratic comparison, symmetry-preserving operators. In the later sections, they were analytical: recurrence, refinement, rotation, summation, and arc-length comparison.

The table is not claimed to be exhaustive. It is claimed to be *ordered*: each constant depends only on structure or completion machinery already on the page, and no constant could have appeared earlier in the ladder than the stage at which it is introduced.

16 Discussion

Scope:

This paper derives a sequence of mathematical constants from the structural operations available to the balanced-ternary state space, but it now makes an explicit distinction between primitive substrate content and later analytical completion demands. Some constants follow once additional structural commitments such as independent generators, quadratic comparison and symmetry-preserving operators are admitted. Others appear only when the substrate is examined through stronger completion principles such as recurrence, refinement, rotation and summation.

The claim is therefore ordered rather than absolute. The paper does not claim that every later constant is already present in raw ternary syntax. It claims that, once each additional demand is stated openly, the resulting constants enter in a strict derivation ladder and do so at the earliest stage permitted by the available machinery. No physical interpretation is assumed or required.

What is new:

The individual constants are, obviously, not new. What is new is the sequential derivation from a single starting point, together with a clear bookkeeping of ontological status. The sequence is not presented as a flat list of consequences all carrying the same force. It is presented as an ordered map from a minimal discrete substrate to the hierarchy of constants revealed when increasingly strong completion demands are imposed.

What is not claimed:

The paper does not claim that all mathematically significant constants can be derived in this way. It does not claim that the derivation chain is the unique such chain. It does not claim that the balanced-ternary substrate, in isolation, is spontaneously "doing calculus" or evaluating global sums without further analytical structure being admitted.

What it does claim is narrower and stronger: given the stated succession of structural and analytical demands, *this particular ladder* is ordered, coherent, and non-arbitrary. Each link follows from machinery already on the page, and the starting point was itself established as a logical necessity in the companion paper.

Open question:

Whether the constants derived here, taken together, suffice to reconstruct a broader mathematical framework beyond the starting alphabet remains open. The present paper does not resolve that question. It establishes only the ladder itself, the order in which its constants appear, and the completion demands under which they do so.

17 Conclusion

The balanced-ternary state space $\{-1, 0, +1\}$ is not inert. Once it is subjected to successive structural and analytical completion demands, it exposes a determinate ladder of constants whose order is not arbitrary. Some arise from comparatively primitive extensions of the substrate, such as independent generators, quadratic comparison and symmetry-preserving operators. Others appear only under stronger analytical completions, including recurrence, refinement, rotation and summation.

The sequence begins with the structure of independent axes and proceeds, in order, through the fundamental irrationals, self-referential growth constants, rotation-based invariants, information-theoretic logarithms, and the simplest convergent integer sums. The paper does not claim that all of these are primitive contents of raw ternary syntax. It claims that, once each added demand is made explicit, the resulting constants enter in a strict sequence at the earliest stage permitted by the available machinery.

No numerical value was tuned. Where a choice of scale is required, it is fixed by an explicit normalisation. No constant is inserted ad hoc. The claim is therefore not that the substrate contains the whole edifice in finished form, but that it supports a disciplined, non-arbitrary derivation ladder whose constants appear in order under its admissible completions.